

CasMLN: Cascaded Multi-Level Learning on Temporal Heterogeneous Graphs with LLM Knowledge

CasMLN (Wang et al., 2024) introduces a unified model that fuses graph learning and LLM-based external knowledge to improve representation learning on **temporal heterogeneous graphs (THGs)**. These are dynamic graphs where both the structure and types of nodes/edges evolve over time, such as user-item interactions, academic citation networks, or COVID case reports.

Multi-Level Learning over Temporal Heterogeneous Graphs (With Examples)

CasMLN captures patterns at three hierarchical levels: node-level, type-level, and graph-level, leveraging both structural and temporal signals.

Example 1: Node-Level Learning

Given a THG snapshot at time t , CasMLN applies relational GCNs:

- Paper **P123** is written by Author **A45** and published at Venue **V2**.
- These edges:
 A45 --write--> **P123**,
 P123 --publish--> **V2**
 allow the GCN to learn structural embeddings by aggregating type-specific neighbor features.

Example Output:

Node embeddings:

- $z_{\text{P123}} = [0.12, 0.09, \dots]$
 - $z_{\text{A45}} = [0.07, 0.11, \dots]$
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Example 2: Type-Level Learning with Degree Centrality

In the Aminer dataset, authors with many co-authored papers (i.e., high degree) are considered more influential.

At time t , for node type **Author**:

- Author **A45** has degree = 10
- Author **A02** has degree = 2

CasMLN applies **softmax over degrees** (with amplification), assigning more weight to high-degree nodes when aggregating into a single "Author-type" representation.

Result:

h_type_Author = weighted sum of {**z_A45**, **z_A02**, ...}

Example 3: Graph-Level Learning with Feature Importance

CasMLN then aggregates the type-level vectors (e.g., Author, Paper, Venue) using a learned sigmoid-based attention mechanism.

If **Author** features are more relevant to the task than **Venue**, the system assigns:

- **weight_Author** = 0.55
- **weight_Paper** = 0.30
- **weight_Venue** = 0.15

Final graph-level vector:

h_graph = 0.55 * **h_type_Author** + 0.30 * **h_type_Paper** + 0.15 * **h_type_Venue**

LLM-Based External Knowledge Enhancement (With Examples)

CasMLN enhances both type-level and graph-level embeddings using summaries generated by an LLM (GPT-3.5), fed through a sentence embedding model (**text-embedding-ada-002**).

Example Prompts:

- Graph prompt:
"Ecomm is a graph where users interact with items via click, buy, cart, and favorite events."
- Node type prompt (User):
"Users engage with items through browsing, purchasing, and favoriting. Clicks imply

interest; purchases imply value perception.”

LLM Output → Embedding:

These summaries are converted into 128-dim vectors and integrated via:

$$h_{\text{enhanced_type}} = h_{\text{type}} \odot \log(|\text{LLM_vector}|)$$

This boosts semantic richness, especially for node types with sparse interaction histories.

Temporal Attention across Snapshots (With Examples)

CasMLN models time using **attention over historical graph snapshots**.

For each node (e.g., Author A45), its embeddings across time steps $\{t-2, t-1, t\}$ are assigned attention weights.

Example:

- At $t-2$: $z_{t-2} = [0.10, 0.08]$, weight = 0.2
- At $t-1$: $z_{t-1} = [0.12, 0.11]$, weight = 0.3
- At t : $z_t = [0.15, 0.13]$, weight = 0.5

Final temporal embedding:

$$z_{\text{final}} = 0.2 * z_{t-2} + 0.3 * z_{t-1} + 0.5 * z_t$$

Evaluation and Results

CasMLN is evaluated on four THG datasets: Aminer, Ecomm, Yelp, and COVID-19. Tasks include:

- Link prediction (e.g., author co-authorship, user-item interaction)
- Node classification (e.g., business category on Yelp)
- Node regression (e.g., daily COVID case counts)

It outperforms baselines like HTGNN, DyHATR, and HGT+ across all tasks.

Ablation studies show:

- Removing LLM knowledge leads to 2–3% drops in performance.
 - Removing multi-level learning causes the largest performance degradation.
 - Temporal attention is critical for modeling recent behaviors.
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Why It Matters for the Survey

CasMLN is the **first model** to apply LLM-based external knowledge to **temporal heterogeneous graphs**, not just static ones. It:

- Bridges structural temporal modeling with semantic language understanding.
- Addresses both **heterogeneity** and **evolution** in dynamic graphs.
- Provides a scalable, interpretable framework using node-type and graph-level reasoning enhanced by LLMs.

Its multi-level attention-driven fusion and the modular integration of LLM signals make it a compelling foundation for future LLM-GNN hybrids on dynamic graphs.